GRADE 1 · BLOCK CODING

Predict, Then Run

Guess where Bit lands, then check

Name: ______________________ Date: ______________________

★ **I can predict where the code lands Bit, then run it to check.**

Read it in your head first

A good coder can read code and PREDICT what it will do before running it. Read the blocks in order in your head, write your prediction, then run the code one block at a time to check.

Bit's number path

0	1	2	3	4	5

Code A

start at 0

forward 2

forward 2

1 Predict, then run Code A

PREDICT: where will Bit land? Write your guess. Then run Code A from 0 to check. Was your prediction right?

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Code B

start at 5

back 1

back 2

2 Predict, then run Code B

PREDICT where Bit will land, write it, then run Code B from 5: back 1, then back 2. What number does Bit land on?

3 You try — change and predict

In Code B, change back 2 to back 1. PREDICT the new landing number first, then run to check. What number now? How did changing that block change where Bit lands?

Big idea

Reading code and predicting what it does helps you catch mistakes before you run it. Then running the code tells you if your prediction was right.

TEACHER GUIDE · GRADE 1 · BLOCK CODING

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Quick facilitation notes & answer key

What this worksheet teaches

Students read instructions and GUESS where Bit will land before running them, then run to check. They also change one block and guess again, describing the effect. Reading and predicting first is how coders check their work. No coding experience needed.

Before you start (no computer needed)

Paper and pencil only. Optional: a floor number line 0 to 5 to check predictions by walking the code.

How to lead it (about 20 minutes)

1. Show them first (you do). Read Code A aloud without moving: "start 0, forward 2 would be 2, forward 2 would be 4 — I predict 4." Then run it to confirm.
2. Do one together (we do). Exercise 1: predict (4), then run Code A. Exercise 2: predict Code B (2), then run.
3. Let them try alone (you do). Exercise 3: change back 2 to back 1; predict (3), run, and describe that a smaller back means Bit lands one higher.
4. Big idea: "Reading and predicting first is how coders check their work before running it."

Answer key (these are the verified correct answers)

Exercise 1 — Code A: 0 → forward 2 → 2 → forward 2 → 4. Predict and land on 4.

Exercise 2 — Code B: 5 → back 1 → 4 → back 2 → 2. Bit lands on 2.

Exercise 3 — change back 2 to back 1: 5 → back 1 → 4 → back 1 → 3. Bit lands on 3 — one higher than before, because the last block now takes away one fewer step.

Why predicting works: the blocks run in a fixed order, so you can trace them in your head — start number, then each move — to know the result before running.

If students get stuck — and ways to adjust

Watch for: predicting by adding all the move numbers and ignoring back blocks. Remind students that back takes away.

Make it easier: trace each code together on the floor line, pausing to predict the next number block by block.

Make it harder: ask students to predict Code A if the start were "start at 1" (lands on 5, the end).

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

In plain terms: students follow (or write) step-by-step instructions, in order, to get a result.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

In plain terms: students read instructions someone wrote, change one, and explain what the change did.

You'll know it worked when

The student writes a correct prediction before running, confirms it, and explains the effect of the changed block.